



List of Departmental Elective - II for Semester-VI:

Sr. No.	Course Code	Course Name	Remarks
1	216U01E611	Compiler Construction	
2	216U01E612	Virtual Reality & Augmented Reality	
3	216U01E613	Embedded System and IOT	
4	216U01E614	Mobile Communication and Mobile App Development	
5	216U01E615	Microservices Foundations	
6	216U01E616	Agile Project Management	
7	216U01E617	Cloud Computing and Virtualization	
8	216U01E618	Machine Learning	Not available to students with Honours in Data Science and Analytics
9	216U01E619	Introduction to Biotechnology & Bioinformatics	



Course Code	Name of the Course			
216U01E617	Cloud Computing & Virtualization			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	03	--	-	03
Credits Assigned	03	--	-	03
Evaluation Scheme	Marks			
	LAB/TUT CA	CA (TH)		ESE
		IA	ISE	
	-	20	30	50
				100

Course pre-requisites:-

1. Fundamental knowledge on Operating system and Computer Networks
2. Basics of client/server programming and network protocols

Course Objectives:

Cloud computing has evolved as a very important computing model, which enables information, software, and other shared resources to be provisioned over the network as services in an on demand manner. Students will be exposed to the current practices in cloud computing. Topics may include distributed computing models and technologies, Infrastructure-as-a-Service (IaaS), Platform-as-a-Service (PaaS), Software-as-a-Service (SaaS), virtualization, performance and systems issues, capacity planning, federated clouds, challenges in implementing clouds, data centers, hypervisor CPU and memory management, cloud hosted applications, and other advanced and research topics in cloud computing.

Course Outcomes (CO):

- CO1. Comprehend the issues related to cloud computing and its application
- CO2. Investigate the system virtualization and outline its role in enabling the cloud computing System model
- CO3. Analyse and apply cloud programming models to solve problems
- CO4. Build cloud services and applications
- CO5. Configure and experiment with advanced cloud technologies

Mod	Unit	Contents	No of	CO
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Sl. No.	No.		Hrs.	
1	Introduction		8	CO1
	1.1	Cloud Computing at a Glance, Historical Developments, Building Cloud Computing Environments, Computing Platforms and Technologies - Microsoft Azure, Hadoop, Force.com and Salesforce.com, Manjrasoft Aneka		
2	Virtualization		10	CO1
	2.1	Introduction, Characteristics of Virtualized Environments , Taxonomy of Virtualization Techniques, Virtualization and Cloud Computing, Pros and Cons of Virtualization CO2		
	2.2	Technology Examples: Xen: Para virtualization, VMware: Full Virtualization, Microsoft Hyper-V		
	2.3	Cloud Computing Architecture : Cloud Reference Model, Types of Clouds, Economics of the Cloud, Open Challenges		
3	Cloud Infrastructure and Platforms in Industry		10	CO2
	3.1	Data-Center Design-Warehouse-Scale Data-Center, Modular Data Center. Interconnection Networks-Fat-Tree Interconnection Topology for Warehouse Scale Data Centers, Server-Centric BCube Network for a Modular Data Center.Data-Center Management Issues		
	3.2	Amazon Web Services – Compute Services, Storage Services, Communication Services, Additional Services		
	3.3	Google Cloud Platform, Google AppEngine: Architecture and Core concepts; Application Life Cycle		
4	Cloud Applications		8	CO2
	4.1	Scientific Applications – Healthcare: ECG analysis in Cloud, Biology: Protein Structure Prediction, Geoscience: Satellite Image Processing		
	4.2	Business and Consumer Applications – CRM and ERP, Productivity, Social Networking, Media Applications, Multiplayer Online Gaming		
		#Self-Learning – other Applications		
5	Advanced Topics in Cloud Computing		9	CO3
	5.1	Energy Efficiency in Clouds, Market Based Management of Clouds, Federated Clouds / Inter Cloud, Third Party		



		Cloud Services: MetaCDN, SpotCloud		
	5.2	Dockers and Containers, Micro Services, Cloud automation tools and DevOps concepts		
Total			45	--